The Multilingual Hispanic Speech in California (MuHSiC) Corpus: A Large-Scale Open-Access Spanish–English Bilingual Corpus

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Abstract

Aims and Objectives / Purpose / Research Questions
This article introduces the Multilingual Hispanic Speech in California (MuHSiC) corpus and addresses two primary questions: (1) how can a large-scale, open-access corpus be designed to capture the linguistic diversity of Spanish–English bilingualism in California, and (2) how can such a resource support reproducible research across linguistic subfields?

Design / Methodology / Approach
MuHSiC was constructed as a sociolinguistically stratified spoken corpus based on systematically collected bilingual interviews. The project integrates high-fidelity audio recording, standardized transcription protocols, phoneme-level forced alignment, acoustic extraction, and structured metadata collection to produce a research-ready, publicly distributable dataset.

Data and Analysis
The corpus comprises 600 recordings from bilingual speakers representing diverse social and regional backgrounds across California. Each recording is approximately 70 minutes in length, evenly divided between Spanish and English, yielding over 42,000 minutes (712 hours) of speech. The dataset contains nearly 4.7 million time-aligned word tokens and over 16 million phonological segments, accompanied by detailed speaker metadata.

Findings / Conclusions
MuHSiC demonstrates that large-scale bilingual speech corpora can be constructed with both

sociolinguistic breadth and phonetic detail. Its design ensures reliable alignment, replicable acoustic analysis, and compatibility with morphosyntactic and discourse-level research.

Originality
To date, MuHSiC is the largest open-access, linguistically annotated corpus of Spanish–English bilingual speech. Its integration of balanced bilingual recordings, phoneme-level segmentation, and large-scale acoustic data distinguishes it from existing corpora.

Significance / Implications
MuHSiC provides an unprecedented empirical foundation for research on bilingual phonology, phonetics, morphosyntax, and language contact. It establishes a methodological benchmark for multilingual corpus construction and supports open, collaborative research on linguistic diversity in California and beyond.

Keywords: Bilingual speech corpora, Spanish-English bilingualism, California, Corpus design, Phonetics; Sociolinguistics

Another version of the abstract (if we don't submit to IJB)

This article introduces the Multilingual Hispanic Speech in California (MuHSiC) project, a large-scale audio and text corpus of bilingual Spanish–English speech. The corpus is based on sociolinguistic interviews with bilingual speakers from diverse social backgrounds and regional origins across California. MuHSiC comprises 600 recordings, each approximately 70 minutes long, evenly divided between English and Spanish, yielding over 42,000 minutes of transcribed and phonologically segmented speech. To date, it is the largest open-access, linguistically annotated corpus of Spanish–English bilingual speech. After situating MuHSiC among comparable corpora, we describe its design and development, including speaker recruitment, metadata collection, elicitation and recording procedures, transcription, phonological annotation, and corpus assembly. MuHSiC provides a valuable resource for investigating phonology, phonetics, morphology, syntax, and lexicon in California's bilingual communities, while establishing a methodological benchmark for corpus-based research in multilingual settings.

1. Introduction

Bilingual speech communities provide crucial insight into processes of language contact, sociolinguistic variation, and change. In contexts where two languages are used in sustained interaction, patterns of phonetic convergence, morphosyntactic transfer, lexical innovation, and discourse-level code-switching emerge in ways that challenge monolingual models of linguistic structure. California, home to one of the largest Spanish-speaking populations in the United States, constitutes a particularly important site for investigating these dynamics. Yet despite the demographic and sociolinguistic significance of Spanish–English bilingualism in the state, large-scale, phonetically robust, and openly accessible corpora of bilingual speech remain scarce.

The Multilingual Hispanic Speech in California (MuHSiC) project addresses this gap. MuHSiC is a large-scale bilingual Spanish–English spoken corpus designed to support research in sociolinguistics, phonetics, phonology, and corpus linguistics. It comprises transcribed text and high-fidelity audio recordings from sociolinguistic interviews and naturalistic conversations with bilingual speakers representing diverse social backgrounds and regional origins across California. The corpus contains 600 recordings, each approximately 70 minutes in length and evenly divided between English and Spanish. In total, MuHSiC includes 712.43 hours of bilingual speech, comprising 4,698,607 time-aligned word tokens and 16,277,689 time-aligned phonological segments. To date, it is the largest open-access, linguistically annotated corpus of Spanish–English bilingual speech and is freely available at https://muhsic.ucsc.edu/.

The corpus was designed to capture the social and geographic diversity of California's bilingual communities. Data collection was coordinated across three University of California campuses (UC Santa Cruz, UC Berkeley, and UCLA) to ensure broad regional coverage and demographic representation. Recordings were structured to elicit naturalistic speech across a range of conversational domains, including migration narratives, family histories, traditional folktales, culinary practices, and metalinguistic reflections on "Spanglish." This topical diversity supports systematic investigation of discourse organization, stylistic differentiation, and dialectal variation in both Spanish and English.

From its inception, MuHSiC was built with phonetic precision as a central priority. All recordings were made using professional-grade head-mounted microphones (AKG C520) and portable recorders (Zoom H4n Pro) in 44.1 kHz, 16-bit, mono, lossless format. These specifications ensure that the corpus supports fine-grained acoustic and phonological analysis while preserving the ecological validity of sociolinguistic interviews. Unlike many general-purpose spoken corpora, MuHSiC integrates high technical recording standards with richly annotated transcripts, enabling research that spans detailed phonetic inquiry and broader studies of bilingual language use.

MuHSiC is accompanied by a searchable interface that allows users to query the corpus by sociolinguistic attributes. The dataset's time-aligned transcription and phonological annotation make it possible to examine correlations among linguistic patterns, geographic origin, and other sociolinguistic and demographic variables. Its open-access design supports research at multiple levels, from undergraduate training to advanced scholarship, while encouraging collaboration between researchers and educators seeking to incorporate authentic bilingual data into empirical and pedagogical work.

This article documents the design and implementation of MuHSiC and proposes it as a methodological model for constructing large-scale bilingual speech corpora optimized for sociolinguistic and phonetic research. Section 2 situates MuHSiC within the landscape of Spanish-language corpora in the United States and outlines its distinctive contributions. Section

3 details the corpus design, including speaker sampling and the structuring of social, geographic, and demographic variables. Section 4 describes data and metadata collection procedures, including recruitment, elicitation, and recording techniques. Section 5 explains the corpus-building workflow, including transcription, anonymization, text extraction, phonological forced alignment, and acoustic analysis. Section 6 outlines access to the corpus and its publicly available components, and Section 7 concludes with implications and future directions for research on bilingualism and language variation.

2. Other Spanish-English bilingual speech corpora

In this section, we review existing Spanish–English bilingual speech corpora in the United States, focusing on their size, linguistic coverage, methodological design, and accessibility.

2.1. CESA

The *Corpus del Español en el Sur de Arizona* (CESA; Carvalho, 2012) consists of 78 sociolinguistic interviews, each approximately 60 minutes long, collected between 2012 and 2018. Participants are native Spanish speakers residing in the Tucson area and represent a range of age groups, genders, and birthplaces. Although interviewers did not constrain code-switching, the interviews were conducted in Spanish, and the corpus is therefore overwhelmingly Spanish-dominant.

CESA was developed to facilitate research on grammatical convergence in the Spanish of Tucson bilinguals and to enable comparisons with Mexican border varieties, particularly Sonoran Spanish. The corpus includes audio files, PDF transcriptions, and spreadsheets containing speaker metadata. While well-suited for morphosyntactic and discourse-level research, CESA does not provide segment-level phonological annotation or standardized high-fidelity recordings designed specifically for acoustic analysis.

2.2. CoBiVa

The *Corpus Bilingüe del Valle* (CoBiVa; Christoffersen & Ciller, 2024) comprises sociolinguistic interviews collected between 2019 and 2024 from bilingual residents of the Rio Grande Valley in southern Texas. Each interview lasts approximately 60 minutes and is conducted primarily in either Spanish or English, though code-switching is not restricted. Speakers include native Spanish speakers, native English speakers, and bilinguals from the region.

CoBiVa aims to document and legitimize local bilingual language norms in the Rio Grande Valley. Audio recordings are accompanied by stream-based transcriptions and questionnaire-based metadata. As with CESA, the corpus offers valuable insight into bilingual language practices but does not systematically provide parallel data in both languages for every speaker, nor does it include force-aligned phonological segmentation for fine-grained phonetic study.

2.3. Spanish in Texas

The Spanish in Texas corpus (Bullock & Toribio, 2024) contains 96 sociolinguistic interviews collected between 2011 and 2013 from Spanish speakers across Texas. Interviews are conducted in Spanish, though code-switching into English is not constrained. The corpus was designed in part as a pedagogical resource, enabling language teachers and learners to engage with Texas Spanish as a legitimate linguistic and cultural variety.

Recordings are available as subtitled video files alongside searchable text transcriptions, part-of-speech annotations, and metadata spreadsheets. While the corpus is searchable and pedagogically oriented, it is primarily Spanish-only and was not constructed with uniform, phonetics-oriented recording protocols suitable for large-scale acoustic analysis.

2.4. Miami Bilingual Conversational Corpus

The Miami Bilingual Conversational Corpus (Deuchar et al., 2014) consists of 56 naturalistic conversations involving 84 participants, recorded between 2008 and 2011. Each conversation lasts at least 30 minutes. Participants are Spanish–English bilinguals residing in Miami and represent diverse ages, genders, and national origins. The corpus is predominantly English (approximately two-thirds English and one-third Spanish), and code-switching occurs freely.

Designed to facilitate research on code-switching in naturalistic interaction, the corpus includes audio files, CHAT-format transcriptions, and participant metadata. Its conversational design makes it particularly valuable for discourse and code-switching studies. However, it does not systematically provide balanced speech samples in both languages for each participant, nor was it designed with standardized high-fidelity recording protocols for phonetic research.

2.5. Justification for MuHSiC

2.5.1. Linguistic justification

Although the corpora reviewed above have made important contributions to the study of Spanish–English bilingualism, MuHSiC diverges from them in several fundamental respects. Most notably, it provides systematically balanced data in both Spanish and English for every participant. Each of its 603 speakers contributes approximately 35 minutes of interview data in each language, yielding 1,203[1] language-specific recordings. In contrast to corpora that rely primarily on spontaneous code-switching within interviews conducted predominantly in a single language, MuHSiC's architecture permits direct within-speaker comparisons across languages. This design substantially strengthens the empirical foundation for examining cross-linguistic influence, structural convergence, and asymmetries between bilingual grammatical systems.

Equally consequential is the fact that MuHSiC was conceived from the outset to support phonetic and phonological inquiry. Beyond orthographic transcription, the corpus includes time-aligned, segment-level phonological annotation derived through forced alignment. Recordings were obtained using a uniform, professional-grade configuration (AKG C520 head-mounted microphones paired with Zoom H4n Pro recorders operating at 44.1 kHz, 16-bit, mono, lossless format), thereby minimizing variability in signal quality across sites and sessions. This degree of technical standardization renders the corpus particularly well suited to fine-grained acoustic analysis, whereas many existing bilingual corpora offer only orthographic transcripts and audio of inconsistent fidelity that is not optimized for phonetic research.

Scale further distinguishes MuHSiC within the landscape of Spanish–English bilingual corpora. With 600 bilingual speakers and more than 700 hours of recorded speech, it substantially exceeds the size of most currently available resources. Such breadth enables

[1] 600 speakers successfully recorded both the English and Spanish portions of the sociolinguistic interviews and metadata entry. Due to poor sound quality or audio corruption, two speakers (UCB_BM1_04_Spanish.wav and UCSC_AT1_09_Spanish.wav) successfully completed only the Spanish portion of the interview, and one speaker successfully completed only the English portion of the interview (UCSC_MB1_08.wav). Due to audio corruption problems, one English file was only recoverable as an mp3 (UCSC_DG1_07_English.mp3, counted as part of the 600 speakers who completed both portions) but the Spanish file UCSC_DG1_07_Spanish.wav remained unchanged. We have reconverted UCSC_DG1_07_English.mp3 into a wav file, recognizing the audio does not have its original quality.

statistically robust modeling of linguistic variation across social, geographic, and demographic dimensions, thereby enhancing the representativeness of analyses of California's bilingual speech communities.

Finally, MuHSiC integrates orthographic transcription, phonological annotation, detailed speaker metadata, and pregenerated acoustic datasets within a unified open-access infrastructure. Rather than constituting a loosely organized repository of audio and text files, it functions as a systematically structured and computationally accessible research platform designed to support inquiry across phonetics, phonology, morphology, syntax, sociolinguistics, and discourse analysis.

2.5.2. Social justification

Beyond its methodological contributions, MuHSiC responds to the broader social context of Spanish–English bilingualism in the United States. Despite the country's large and historically rooted Spanish-speaking population, bilingualism has frequently been stigmatized, and U.S. Spanish has often been characterized as deficient or "mixed" (ANLE, 2010). Such ideologies have influenced educational policy, public discourse, and speakers' own linguistic self-perception.

California, which has the largest Hispanic population in the United States (Hugo Lopez & Manuel Krogstad, 2015; Johnson, Cuellar Mejia, & McGhee, 2026; World Population Review, 2026), represents a particularly important site for examining contemporary bilingual practices. Earlier research has documented patterns of intergenerational language shift and the gradual reduction of Spanish use in immigrant communities. At the same time, bilingual speakers, especially younger generations, are often described in both public and academic discourse as producing a simplified or substandard variety shaped by English influence (ANLE, 2010; Cherry, 2011; Goble, 2016; Higby, Gámez, & Holguín Mendoza, 2023; Otheguy, 2016; Prada, Guerrero-Rodriguez, & Pascual, 2020; Tseng, 2021). These characterizations risk obscuring the structural complexity and systematicity of bilingual grammars.

A large-scale, linguistically robust corpus such as MuHSiC enables the empirical examination of these claims. By documenting Spanish and English as used by bilingual speakers across the state, the corpus provides evidence for understanding language contact, variation, and change without presupposing deficit-based assumptions. In doing so, MuHSiC contributes not only to linguistic theory but also to the broader effort to legitimize U.S. Spanish and to challenge persistent forms of linguistic prejudice.

3. Corpus design and speaker sampling

MuHSiC consists of sociolinguistic interviews with Spanish–English bilingual speakers residing in California. Its design reflects two central priorities: (i) to obtain a large, socially and geographically diverse sample of bilingual speech representative of California's multilingual communities, and (ii) to facilitate research not only in corpus linguistics but also in sociolinguistics, phonetics, phonology, morphosyntax, and discourse analysis. This section describes the structure of the speaker sample and the criteria used to include participants in the corpus.

3.1. Spanish–English bilingual speaker sample

The principal aim of MuHSiC is to document both Spanish and English as spoken by bilinguals across California. MuHSiC systematically elicits extended speech in each language from every

participant. Each speaker contributed approximately 35 minutes of recorded speech in English and 35 minutes in Spanish. While code-switching was not restricted during interviews, the structured dual-language format ensures that substantial monolingual mode data are available for both languages for all speakers.

This design enables direct within-speaker comparison across languages, permitting fine-grained analyses of bilingual phonetic convergence, phonological patterning, morphosyntactic transfer, lexical choice, discourse organization, and pragmatic strategies. Because all speakers contribute comparable quantities of speech in each language, MuHSiC supports large-scale quantitative studies of bilingual grammars in ways that corpora based solely on spontaneous language mixing cannot.

Speaker recruitment was coordinated across multiple regions of California in order to maximize geographic coverage and demographic diversity. Interviews were conducted through coordinated fieldwork efforts centered at three University of California campuses (UC Santa Cruz, UC Berkeley, and UCLA). Participants represent varied social backgrounds, migration histories, and regional affiliations within the state. Figure 1 illustrates the primary residence locations of speakers across California.

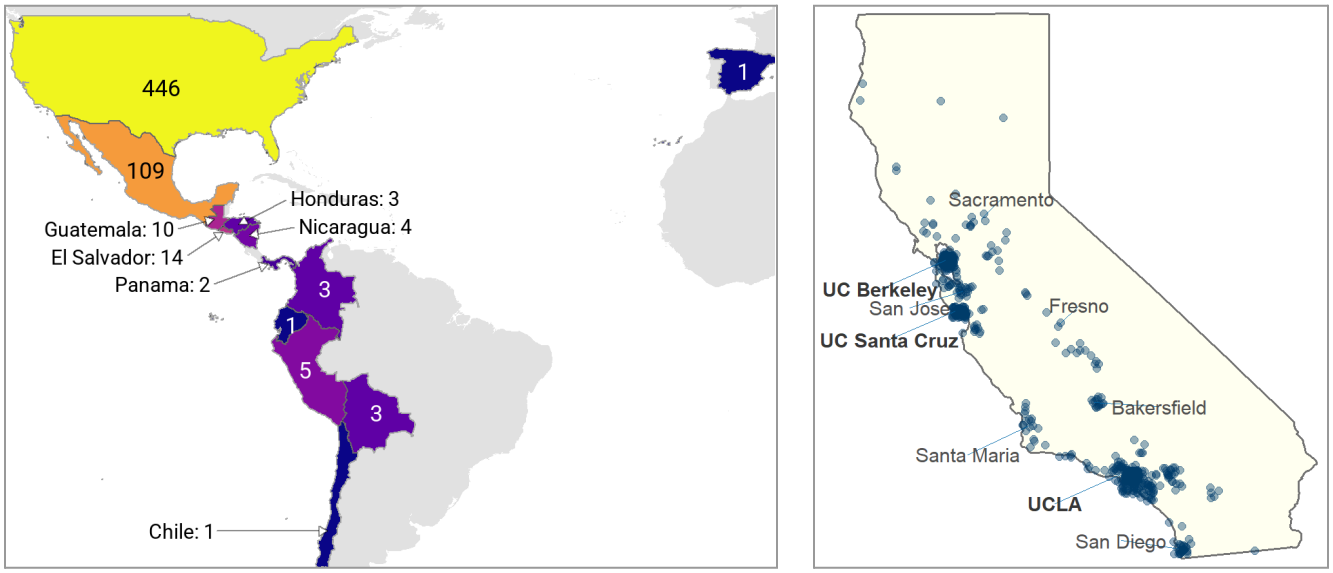

Figure #. Left: Number of MuHSiC interviewees by country of birth. Right: A map of the interviewee's primary[2] location within California.

3.2. Speaker inclusion criteria

Clear and consistent inclusion criteria were essential to ensuring comparability across speakers and localities. Participants were required to meet the following conditions: (i) all speakers had to be long-term residents of California. A minimum residency of five consecutive years in the state was required to ensure meaningful exposure to local linguistic norms and participation in California's bilingual speech communities. Also, (ii) participants were required to demonstrate

[2] If two participants list the same location (eg., several points in Los Angeles), the map slightly offsets each point so that there isn't so much overlap that it becomes entirely illegible. This highlights readability in urban areas and accuracy in rural areas. Primary Location means that 'city growing up' is first reflected in the graph if the participant listed a city in California. If data about 'city growing up' wasn't listed, or if 'city growing up' was from outside of California, then city of current residence was listed. Only three interviewees didn't list a city in the questionnaire, leaving location questions blank.

conversational proficiency in both Spanish and English. Because the corpus is designed to support systematic cross-linguistic comparison, functional ability in both languages was necessary to complete the dual-language interview format.

At this stage of development, MuHSiC only includes recordings of multilinguals producing Spanish and English speech. Bilingual speakers who use Spanish and English almost exclusively were prioritized in participant recruitment to maintain analytic clarity in isolating Spanish–English contact phenomena. Of the 603 interviewees, 40 participants self-identified as speaking a third language without specifying their level of proficiency[3]. Future expansions of the corpus aim to better incorporate these populations to more fully reflect the state's multilingual diversity.

Participant age was not controlled for during the speaker recruitment process. Nevertheless, participants tended to fall within a typical university student age range; 2003 was the most common birth year during the May 2023 to August 2025 recording sessions.

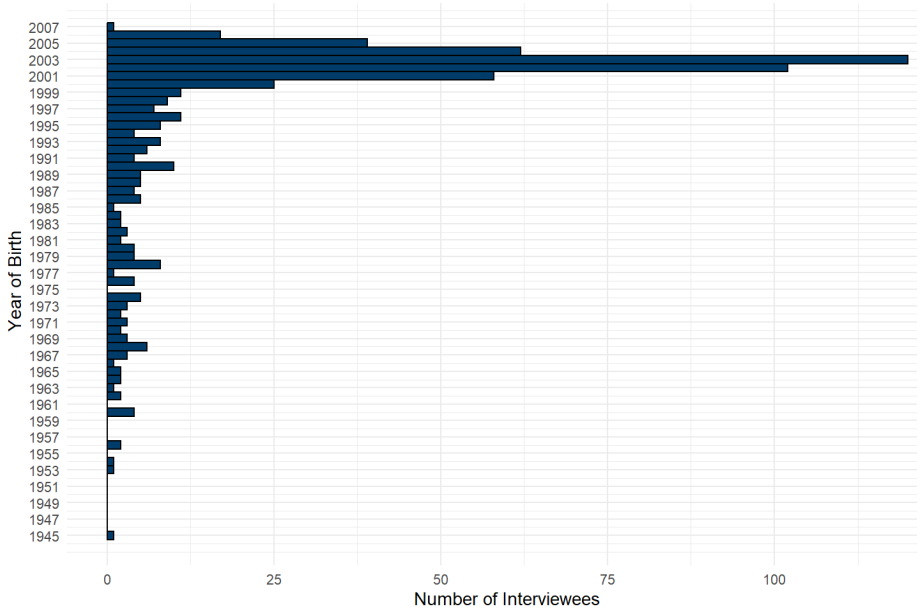

Figure #. Years of birth of interviewees in the corpus.

3.3. Design priorities and research flexibility

Several decisions were made to enhance the corpus's representativeness and analytical flexibility. Crucially, detailed socio-demographic metadata were collected from all participants. These metadata enable researchers to investigate correlations between linguistic variables and social factors such as age, gender, migration background, education, and regional affiliation.

In addition, interviews were structured to elicit naturalistic yet comparable speech across participants. A consistent interview framework, combined with open-ended narrative prompts, ensures both ecological validity and cross-speaker comparability. The dual-language format of the corpus enables the analysis of bilingual speech along a continuum ranging from predominantly single-language production to spontaneous code-switching. This flexibility

[3] A handful of speakers listed that they spoke 'a little' of a language, or 'elementary' proficiency, and were not included in this count of 40 speakers.

accommodates research traditions that prioritize controlled comparisons as well as those that foreground naturalistic bilingual practices.

The corpus was built with phonetic research in mind. High-fidelity recording protocols and consistent equipment were used across sites to minimize variation in recording conditions. This allows for reliable acoustic analysis across speakers, regions, and social groups. These features make MuHSiC a large-scale, socially grounded, and methodologically rigorous resource for the study of Spanish–English bilingualism in California.

4. Data collection

This section details the procedures used to collect the MuHSiC corpus. After specifying the timeframe of data collection, we describe the selection and training of fieldworkers, speaker recruitment and exclusion criteria, interview protocols, elicitation design, and recording procedures.

4.1. Timeframe

MuHSiC recordings were collected between May 2023 and August 2025 across multiple locations throughout California. Data collection proceeded concurrently at three University of California campuses (UC Santa Cruz, UC Berkeley, and UCLA), allowing for broad geographic coverage and efficient recruitment of participants from diverse regional backgrounds.

4.2. Fieldworkers

Fieldworkers were responsible for recruiting eligible speakers, obtaining informed consent, administering questionnaires, conducting and recording interviews, and submitting complete documentation for each session.

A total of 75 fieldworkers participated in the project. These included 18 undergraduate students from UC Santa Cruz, 36 undergraduate students from UC Berkeley, and 21 students from UCLA (20 undergraduate, 1 graduate). The majority were native Spanish–English bilinguals (UCSC: 14; UCB: 28; UCLA: 19), while a smaller number were advanced L2 speakers of Spanish (UCSC: 4; UCB: 8; UCLA: 2). All fieldworkers were fluent in both Spanish and English and were long-term California residents, either born and raised in the state or having spent most of their lives there.

Prior to conducting interviews, fieldworkers completed structured training sessions covering ethical procedures, interview techniques, equipment configuration, and file management. This training minimized inter-fieldworker variability and promoted methodological standardization across sites.

Each fieldworker received a standardized field kit containing: (i) a Zoom H4n Pro recorder with a 32GB SD card; (ii) an AKG C520 head-mounted microphone; (iii) a USB cable or SD card reader; (iv) rechargeable AA batteries and charger; (v) participant compensation ($25 gift cards); and (vi) printed materials, including the interviewer manual, consent forms, interview prompts for the Spanish and English sessions, a socio-demographic questionnaire, the Bilingual Language Profile (BLP) questionnaire (Birdsong, Gertken, & Amengual, 2012), payment log, and field notes template. The use of identical equipment and standardized materials across all sites was essential to ensuring recording consistency and data comparability.

4.3. Speaker recruitment

4.3.1. Recruitment procedures

Speakers were recruited primarily through fieldworkers' social networks on campus and in their hometowns, including family members, friends, and community contacts. This approach facilitated access to diverse bilingual speakers across different regions of California.

Potential participants first completed informed consent procedures (Section 4.4) and a socio-demographic questionnaire to verify that they met the inclusion criteria described in Section 3.2. Only speakers who satisfied all eligibility requirements were recorded.

4.3.2. Exclusion criteria

In addition to the inclusion criteria outlined in Section 3, further exclusion parameters were established to ensure the integrity and analytical value of the dataset. Recordings were removed from the corpus if they contained extended periods of silence or if the participant contributed substantially less speech than the interviewer, thereby limiting their usefulness for quantitative linguistic analysis. Sessions were likewise excluded when significant interruptions or sustained audible disruptions, such as third-party intrusions or prolonged environmental noise, compromised the phonetic quality, continuity, or naturalness of the interaction. Finally, interviews were discarded if recording quality proved inadequate because of improper microphone placement, incorrect gain settings, or technical malfunction that materially affected the clarity of the speech signal.

The application of these criteria ensured that the final corpus met rigorous standards for audio fidelity and participant contribution, thereby preserving its suitability for detailed phonetic, phonological, and sociolinguistic investigation.

4.4. Interview procedures

Each session lasted approximately 90 minutes and was conducted in a quiet, furnished room with only the fieldworker and participant present. Fieldworkers were instructed to minimize background noise by closing doors and windows, turning off electronic devices (e.g., fans, air conditioners), and pausing recording if external noise became intrusive. Participants were asked to silence or turn off mobile phones prior to recording.

The session proceeded in the following order: (i) informed consent and recording setup; (ii) English sociolinguistic interview; (iii) written socio-demographic questionnaire (in Spanish); (iv) Spanish sociolinguistic interview; (v) Bilingual Language Profile questionnaire (in either English or Spanish); (vi) compensation; and (vii) completion of field notes by the interviewer.

This ordering was designed to ease participants into the recording situation while facilitating a structured language-mode shift between English and Spanish.

Participants read and signed an informed consent form prior to recording. During this time, fieldworkers configured the recording equipment.

The Zoom H4n Pro recorder was set to multitrack mode (MTR) using a single external input. The AKG C520 head-mounted microphone was connected to input port 1. Audio was recorded in WAV format at 44.1 kHz sampling rate, 16-bit resolution, mono channel. Each participant was assigned a unique alphanumeric identifier consisting of campus code, fieldworker initials and number (for cases where multiple fieldworkers shared initials), and participation order (e.g., UCSC_MA1_01). This identifier replaced personal names in all files to ensure confidentiality. The microphone was positioned approximately two inches from the participant's mouth to maintain a consistent signal-to-noise ratio across recordings.

4.5. Sociolinguistic interviews

Each participant completed two conversational interviews: one in English and one in Spanish. The English session was conducted first. Each lasted approximately 35 minutes. Fieldworkers selected prompts from a standardized list of 40 open-ended topics covering childhood memories, family history, migration experiences, daily routines, media consumption, food, and experiences growing up bilingual in the United States. Although thematic overlap existed across languages, prompts were not identical, reducing repetition while maintaining comparability. Fieldworkers were instructed to avoid reading questions verbatim and to foster a natural conversational atmosphere. They were trained to use follow-up questions, use backchanneling strategies that do not interrupt the interviewee, and strategic silence to encourage extended discourse while minimizing interviewer speech. The goal of these interviews was to elicit spontaneous, naturalistic speech suitable for sociolinguistic, discourse, and phonetic analysis.

4.6. Socio-demographic questionnaire

Between the two interviews, participants completed a written socio-demographic questionnaire in Spanish. This questionnaire collected detailed information regarding place of birth, residence history, family language background, education, travel, and language use patterns. Administering the questionnaire in Spanish served a dual purpose: it collected critical metadata and facilitated a shift into Spanish-language mode before the second interview.

4.7. Bilingual Language Profile (BLP)

Following the Spanish interview, participants completed the Bilingual Language Profile (BLP; Birdsong et al., 2012) in the language of their choice. The BLP is a user-friendly, open-source instrument designed to assess language dominance and has been supported and distributed since 2012 by the Center for Open Educational Resources and Language Learning (COERLL) at the University of Texas at Austin. It provides a comprehensive and multidimensional profile of bilingual experience based on self-reported data (Gertken et al., 2014).

The instrument consists of 19 questions distributed across four modules: Language History, Language Use, Language Proficiency, and Language Attitudes. Participants respond separately for each language. The Language History module gathers information on age of acquisition, age of comfort in each language, years of formal education, time spent residing in regions where each language is spoken, and professional experience requiring each language. The Language Use module assesses the proportion of weekly language use across social domains (e.g., family, friends, school, work) and internal speech patterns such as counting or talking to oneself. In the Language Proficiency module, participants rate their speaking, listening, reading, and writing abilities on a scale from 0 ("not well at all") to 6 ("very well"). Finally, the Language Attitudes module probes perceived authenticity, cultural identification, and the importance placed on speaking or being perceived as a native speaker of each language.

Responses generate separate scores for each language within each module, and global scores for each language across modules. An overall dominance score is calculated by subtracting one language's score from the other, yielding a continuous index ranging from –218 (maximally Spanish-dominant) to +218 (maximally English-dominant), with values near zero indicating balanced bilingualism. These indices provide a flexible, quantitative measure of relative dominance that can be used to classify participants or incorporated as a predictor variable in statistical analyses of linguistic outcomes (Amengual, 2026).

4.8. Compensation and field notes

Upon completion of the session, participants received a $25 gift card as compensation and signed the payment log. After the participant departed, fieldworkers completed a structured field note documenting contextual information not captured in the audio recording, including the participant's demeanor, degree of comfort, relationship to the interviewer, and any situational factors potentially relevant to interpretation of the data.

4.9. Recording standards

Because MuHSiC was designed to support phonetic and phonological research, careful attention was given to recording consistency. All sessions used identical head-mounted microphones and digital recorders, with fixed sampling rate and bit depth. The constant microphone-to-mouth distance ensured stable amplitude and reduced variability from speaker movement.

Recordings underwent minimal post-processing, limited to file organization and secure storage. With the exception of personally identifiable information being redacted from the recordings, the original audio files are available without filtering, compression, or signal-altering processing, preserving the acoustic integrity of the recordings. Audio files are also available in 16 kHz downsampled format and compressed mp3 format.

5. Transcription, text processing, and corpus compilation

Transforming the MuHSiC recordings and associated metadata into a research-ready corpus involved several stages: automated transcription, manual transcription correction, metadata standardization, anonymization, generation of time-aligned annotation files, phoneme-level forced alignment, large-scale acoustic extraction, and compilation of derived datasets. Each stage is detailed below.

5.1. Transcription

All recordings were first transcribed automatically using Sonix.ai, a cloud-based speech recognition platform.

Because automated transcription alone does not provide sufficient accuracy for linguistic research, all transcripts were manually corrected by trained undergraduate research assistants at UC Santa Cruz, UC Berkeley, and UCLA. Transcribers corrected lexical errors, punctuation, segmentation, code-switching boundaries, and non-lexical items (e.g., discourse markers, hesitations). To ensure consistency across sites, assistants followed a standardized transcription protocol specifying orthographic conventions, treatment of false starts and truncations, representation of laughter and other paralinguistic material, and formatting guidelines.

After this first round of correction, all transcripts were independently reviewed and corrected by the fourth author for consistency, completeness, and adherence to protocol. This secondary review ensured uniform formatting and minimized inter-transcriber variability.

Due to the use of head-mounted microphones worn by participants, interviewer speech is significantly quieter than interviewee speech. Because interviewer contributions are both acoustically suboptimal and analytically redundant across interviews, only interviewee speech was retained for phonetic alignment and acoustic analysis. The resulting corpus therefore consists exclusively of participant speech, and speaker attribution at all annotated timepoints is unambiguous.

Final orthographic transcripts were exported in DOCX, PDF, and TXT formats to facilitate research in morphosyntax, discourse analysis, and other text-based domains.

5.2. Anonymization and protection of speaker privacy

Speaker confidentiality was prioritized throughout corpus preparation. Personally identifiable information (PII), including full names, addresses, phone numbers, and other identifying references, was manually tagged during transcript correction and subsequently redacted from all publicly distributed text files.

To further protect participants, timestamps corresponding to utterances containing tagged PII were removed from the publicly available WAV files and replaced with a silent interval of equal duration to the original utterance. This prevents reconstruction of redacted content from the audio signal while preserving the integrity of the remaining recording.

All project personnel signed confidentiality agreements, and speaker identifiers were replaced with anonymized alphanumeric codes in all files and derived datasets. Non-redacted audio files and mapping between identifiers and personal information are stored separately in encrypted repositories and are not distributed.

5.3. TextGrid generation and phoneme-level alignment

For time-aligned annotation, corrected transcripts were exported from Sonix as SubRip (.srt) subtitle files containing utterance-level timestamps and speaker labels. These files were programmatically converted into single-tier utterance-level TextGrid files using a custom Python script.

Because interviewer speech is acoustically attenuated and unsuitable for reliable forced alignment, only the interviewee tier was retained in the final TextGrids. This ensures both acoustic consistency and alignment accuracy.

All utterances were subsequently force-aligned at the phoneme level using the Montreal Forced Aligner (McAuliffe et al., 2017). Alignment employed the English (U.S., ARPABET-based) and Spanish (IPA-based) acoustic models and pronunciation dictionaries. Given the finite coverage of the default lexicons, high-frequency out-of-vocabulary items (≥50 tokens) were manually assigned phonemic transcriptions prior to alignment to improve alignment stability and coverage.

The resulting TextGrids contain utterance-level and phoneme-level segmentation suitable for phonetic, phonological, and prosodic analysis.

5.4. Acoustic extraction

To facilitate large-scale phonetic research, acoustic measurements were automatically extracted from all force-aligned recordings using the Praat Mass Analyzer script (Vargo 2026; github.com/julian-vargo/Praat-Mass-Analyzer). The resulting dataset comprises approximately 16 million phoneme tokens, each represented as a single row. For every segment, the database includes phoneme identity, the immediately preceding and following phonemes, the orthographic word in which the segment occurs, and the corresponding speaker identifier. Temporal information is provided in the form of phoneme onset and offset timestamps.

In addition to segmental and contextual information, a comprehensive set of acoustic measurements is included, using the default parameters of the Praat Mass Analyzer. These consist of formant frequencies (F1–F5), pitch (F0), formant bandwidths, harmonicity, and multiple measures of intensity, including mean intensity, intensity extrema, and the difference in intensity relative to the preceding segment. Perturbation measures such as jitter and shimmer are also provided, along with the first four spectral moments and A1–P0. Where applicable, acoustic

parameters were sampled at 10% intervals across the duration of each phoneme in order to capture within-segment dynamics and fine-grained temporal variation.

All analyses were conducted using the default configuration of the publicly available script to ensure transparency and replicability. The final acoustic dataset is distributed in three formats: comma-separated values (CSV), Parquet, and Feather. The latter two binary formats facilitate efficient storage and rapid computational processing of this large-scale dataset (Richardson et al., 2026).

5.5. Corpus compilation

The final corpus consists of multiple integrated components designed to support a wide range of linguistic analyses. It includes anonymized orthographic transcripts in both DOCX and PDF formats, as well as utterance- and phoneme-aligned TextGrid files that allow for precise temporal analysis of the recordings. The audio component is provided in the form of redacted WAV files, ensuring speaker confidentiality while preserving high acoustic quality for research purposes.

Additionally, the corpus incorporates detailed speaker metadata derived from socio-demographic questionnaires and Bilingual Language Profile (BLP) scores, enabling researchers to examine linguistic patterns in relation to social, geographic, and language dominance variables. The pregenerated phoneme-level acoustic dataset further supports large-scale phonetic and phonological investigation.

All files are systematically named using standardized speaker identifiers and organized within a consistent directory structure to ensure transparency, reproducibility, and compatibility with large-scale computational workflows. Taken together, these materials constitute an integrated and versatile resource for research in bilingualism, sociolinguistics, phonetics, phonology, and discourse analysis.

6. Metadata Compilation

Several

Some or completed graduate education	21
Some or completed college	391
Some or completed high school	143
Some or completed primary school	13
Unknown/Not listed	35

Table #. Self-reported level of education of the interviewees in the corpus.

- data table
- metadata summary
- bilingualism-specific variables, generation
- L1,L2,
- Age groups (decide whether to bin or use age groups)
- binary language dominance, and then plot continuous language dominance
- compare dominance by generation number

		English subcorpus	Spanish subcorpus	Combined
Words	***Mean***	4523.46	3302.41	3912.93
	Std. Dev.	1183.07	896.13	1213.87
	Total	2727644	1991351	4718995
Phones	***Mean***	14358.58	12754.22	13556.40
	Std. Dev.	3726.74	3508.85	3705.87
	Total	8658225	7690792	16349017
Audio Duration (sec)	***Mean***	2144.31	2132.53	2138.42
	Std. Dev.	123.67	138.56	131.41
	Total	1,286,856	1285914	2578930

Table #. Word counts, phone counts, and audio counts for the English portion, Spanish portion, and combined portions of the MuHSiC Corpus.

7. Availability and access

The MuHSiC corpus is freely available at https://muhsic.ucsc.edu. All publicly distributable components of the corpus may be downloaded directly from the website without subscription fees. Users may access the materials either as compressed ZIP archives (for WAV, MP3, TextGrid, CSV, DOCX, and PDF files) or as standalone files in high-efficiency binary formats (Parquet and Feather), which are optimized for large-scale computational analysis.

The downloadable corpus includes: (i) anonymized orthographic transcripts in DOCX and PDF formats; (ii) utterance- and phoneme-aligned TextGrid files; (iii) redacted WAV and MP3 audio files; (iv) a pregenerated phoneme-level acoustic dataset containing approximately 16 million observations; and (v) accompanying speaker metadata files containing social, geographic, and demographic variables. All files follow a standardized naming convention and directory structure to facilitate reproducibility and programmatic access.

To protect participant confidentiality, utterances containing personally identifiable information have been redacted from the publicly distributed audio files, and all transcripts have undergone anonymization. Unredacted audio and the mapping between speaker codes and identifying information are stored separately in encrypted form and are not publicly accessible. The MuHSiC website also provides documentation describing transcription conventions, alignment procedures, acoustic extraction parameters, and metadata coding practices. By making both the data and its governing protocols publicly available, the project aims to ensure transparency, replicability, and long-term sustainability.

7. Conclusions and future directions

As the largest open-access, linguistically annotated corpus of Spanish–English bilingual speech to date, MuHSiC provides an unprecedented empirical foundation for the study of bilingualism in California. Its scale, geographic breadth, phonetic resolution, and integration of rich sociolinguistic metadata enable lines of inquiry that were previously impractical or impossible using smaller, locally restricted, or unevenly annotated datasets.

The corpus makes possible large-scale investigations of phonetic variation across bilingual communities, fine-grained analyses of phonological systems in contact, and quantitative modeling of morphosyntactic and discourse-level phenomena in both Spanish and English. Because recordings are balanced across languages and accompanied by detailed

demographic and language background information, researchers can examine correlations between linguistic patterns and factors such as regional origin, migration history, language dominance, and generational status. Moreover, the inclusion of a high-resolution, pregenerated acoustic dataset substantially reduces the time and technical barriers typically associated with phonetic research, enabling analyses across millions of segments without requiring users to perform alignment and measurement independently.

MuHSiC responds to a long-recognized need in research on the phonetics of bilingualism: the need for large-scale naturalistic speech data to complement findings derived from laboratory-based and controlled experimental studies. Experimental work in California has yielded important insights into Spanish–English bilingual phonetics, phonology, and processing (Amengual, 2016, 2018, 2019; Barlow, 2014; Kim et al., 2025; Kim, 2020; Kim & Repiso-Puigdelliura, 2020; Repiso-Puigdelliura, 2024; Repiso-Puigdelliura et al., 2021; Repiso-Puigdelliura & Kim, 2021), but it necessarily elicits speech under structured conditions that may heighten monitoring, reduce stylistic variation, or limit the range of interactional phenomena observed. By contrast, MuHSiC is built from extended sociolinguistic interviews designed to maximize comfort, spontaneity, and conversational depth. The resulting recordings capture bilingual speech as it is used in everyday contexts, thereby allowing researchers to test whether patterns observed in laboratory settings are replicated, amplified, or attenuated in natural discourse.

Beyond its research applications, MuHSiC supports pedagogical innovation. The corpus can serve as a foundation for corpus-informed language instruction, sociolinguistics coursework, undergraduate research training, and graduate-level phonetic analysis. Its open-access design ensures that institutions with limited resources can participate in rigorous bilingualism research without the financial and logistical burdens of large-scale fieldwork.

The next phase of MuHSiC's development will focus on expansion and enrichment. As funding permits, we plan to increase the number of speakers and geographic sites represented, incorporate longitudinal components where feasible, and refine the metadata inventory to include more detailed measures of language experience and proficiency. We also aim to improve alignment accuracy through iterative pronunciation dictionary expansion and to develop an interactive web-based query interface that will allow users to conduct complex searches without specialized programming expertise.

Because multilingualism in California is dynamic and continually evolving, MuHSiC is not intended as a static archive but as an ongoing collaborative enterprise. We therefore invite partnerships with institutions across California and beyond that share the goal of documenting bilingual speech communities with methodological rigor and open scientific principles. Through sustained growth and collaboration, MuHSiC seeks to establish a long-term empirical record of Spanish–English bilingualism in California and to provide a replicable model for large-scale multilingual corpus construction.

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ADD REFERENCES THAT ARE MISSING HERE

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Justin Davidson is an Associate Professor of Hispanic and Romance Linguistics in the Department of Spanish and Portuguese at the University of California, Berkeley. A sociolinguist, his research explores language variation and change in multilingual Spanish-speaking populations (Catalan-Spanish, English-Spanish, Quechua-Spanish) as evidenced from production patterns and language ideologies. Using empirical methods, his work serves to document and legitimize multilingual speech as systematic and non-aberrant.

Ji Young Kim is an Associate Professor in the Department of Spanish and Portuguese at the University California, Los Angeles. She received her Ph.D. in Spanish linguistics at the University of Illinois at Urbana-Champaign. Her research focuses on the production and perception of heritage and second language speech sounds.

Julian Vargo is a Ph.D. student in Hispanic Linguistics at the University of California, Berkeley. His research focuses on acoustic phonetic variation cross-linguistically. Working with languages such as Spanish, Ladino, Catalan, Romanian, and Kalaallisut, he researches how suprasegmental vowel acoustics provide insight into phonological and socio-dialectological patterns.

Appendix

Italian: 6	Quechua: 2

French: 4	Dutch: 1
Portuguese: 4	Japanese: 1
German: 3	Korean: 1
Mixtec: 3	Mam: 1
Zapotec: 3	Mandarin: 1
American Sign Lang.: 2	Maya: 1
Cantonese: 2	Not listed (wrote "yes" to L3): 3
Nahuatl: 2	

Table #. Self-reported third languages across the 40 identified participants.